



Government College Of Engineering Kalahandi

Presents'

HACKVERSE

FIRST EDITION 2026



TEAM NAME-CODE ALPHA

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***PS ID-OFFLINE-FIRST DIGITAL LEARNING APP
FOR UNDER RESOURCED SCHOOLS***





Team Members

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— The Problem

The Digital Learning Gap in Under-Resourced Schools

Despite the potential of digital education, many students in under-resourced schools are left behind.

01

Limited Internet Access

Unreliable or expensive connectivity makes online learning difficult.

02

Lack of Digital Resources

Many schools have limited access to quality educational content and devices.

03

Teacher Constraints

Teachers often lack tools to provide personalized learning at scale.

04

One-Size-Fits-All Learning

Students with different learning levels often receive the same content.

05

Low Engagement

Traditional learning methods can make it difficult for students to stay motivated.

Core Problem

How can we provide **engaging, personalized, and quality digital education** — even where reliable internet access is unavailable?



WHY OUR OFFLINE LEARNING APP IS DIFFERENT

Built for Under-Resourced Schools. Solves Real Problems. Delivers Real Learning.



WHAT UNDER-RESOURCED SCHOOLS NEED

- ✓ More textbook material
- ✓ Quality, curriculum-aligned content
- ✓ Easy to use on devices
- ✓ Support for teachers
- ✓ Engaging learning for students
- ✓ Low or no recurring cost
- ✓ Local language & local context



OUR OFFLINE LEARNING APP

- ✓ 100% offline – no internet required
- ✓ Interactive, gamified & curriculum-aligned lessons
- ✓ Run smoothly on low-end Android devices
- ✓ Built-in teacher tools (lesson plans, assessments)
- ✓ Engaging, age- & needs-based learning (videos, images, quizzes)
- ✓ One-time setup, no recurring cost
- ✓ Supports local language and local content



OTHER ONLINE LEARNING PLATFORMS

- ✗ Needs constant and reliable internet
- ✗ Limited or no access when offline
- ✗ Subscription-based, recurring cost
- ✗ Often require high-spec devices
- ✗ Limited support for teachers
- ✗ Content not always localized



TRADITIONAL METHODS

- ✗ No digital access
- ✗ Fixed textbooks, library or teacher
- ✗ High cost of printed materials
- ✗ No interactive or adaptive learning
- ✗ No real-time progress tracking
- ✗ Hard to update and not localized



THE PROBLEM

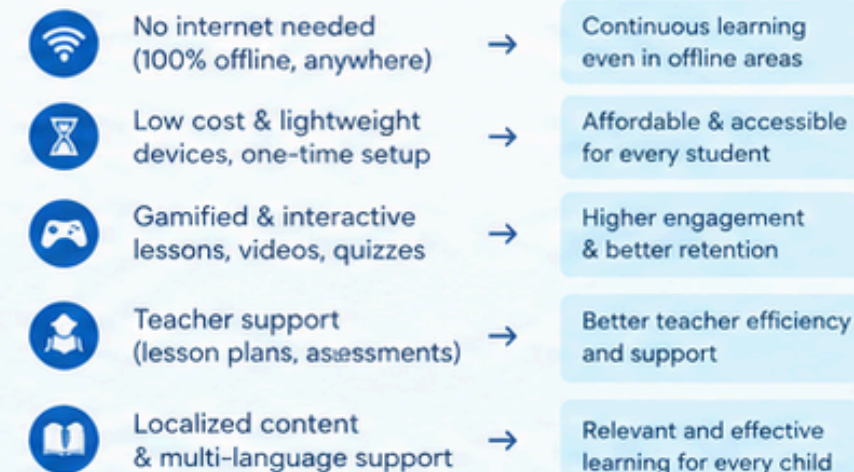
- ✗ Many under-resourced schools have little or no internet access.
- ✗ Digital solutions are often expensive and require high-end devices.
- ✗ Students get limited, non-interactive learning with low engagement.
- ✗ Teachers lack easy-to-use tools and support.



OUR SOLUTION: OFFLINE LEARNING APP



HOW IT SOLVES THE PROBLEM



OUR UNIQUE STRENGTHS

- ✓ **100% Offline-First** (No internet. No problem.)
- ✓ **Gamified Learning Experience** (Makes learning fun and engaging.)
- ✓ **Built for Under-Resourced Schools** (Low cost, high impact.)
- ✓ **Teacher-Centric Design** (Simplifies teaching with tools, plans and progress.)
- ✓ **Local and Sustainable** Supports local language & curriculum. No recurring costs.



OUR OFFLINE LEARNING APP ENSURES **EQUAL ACCESS** TO QUALITY EDUCATION — **NO MATTER WHERE A CHILD IS.**



1. LEARNING FEATURES

- Intercative lessons
- video/ audio learning
- Downloadable study materials
- Practice exercises

2. ENGAGEMENT FEATURES

- Gamified learning
- Progress tracking
- leaderboards



4. SMART FEATURES

- AI- powered learning assistance
- Doubt - solving support
- Adaptive quizzes
- Student can ask using voice

3. MANAGEMENT FEATURES

- Student profiles
- Perfomance analytics
- Sync data when internet becomes available

5. TEACHER FEATURES

- Teacher dashboard
- Student performance tracking
- Quiz creation
- Learning progress reports

6. ACCESSIBILITY FEATURES

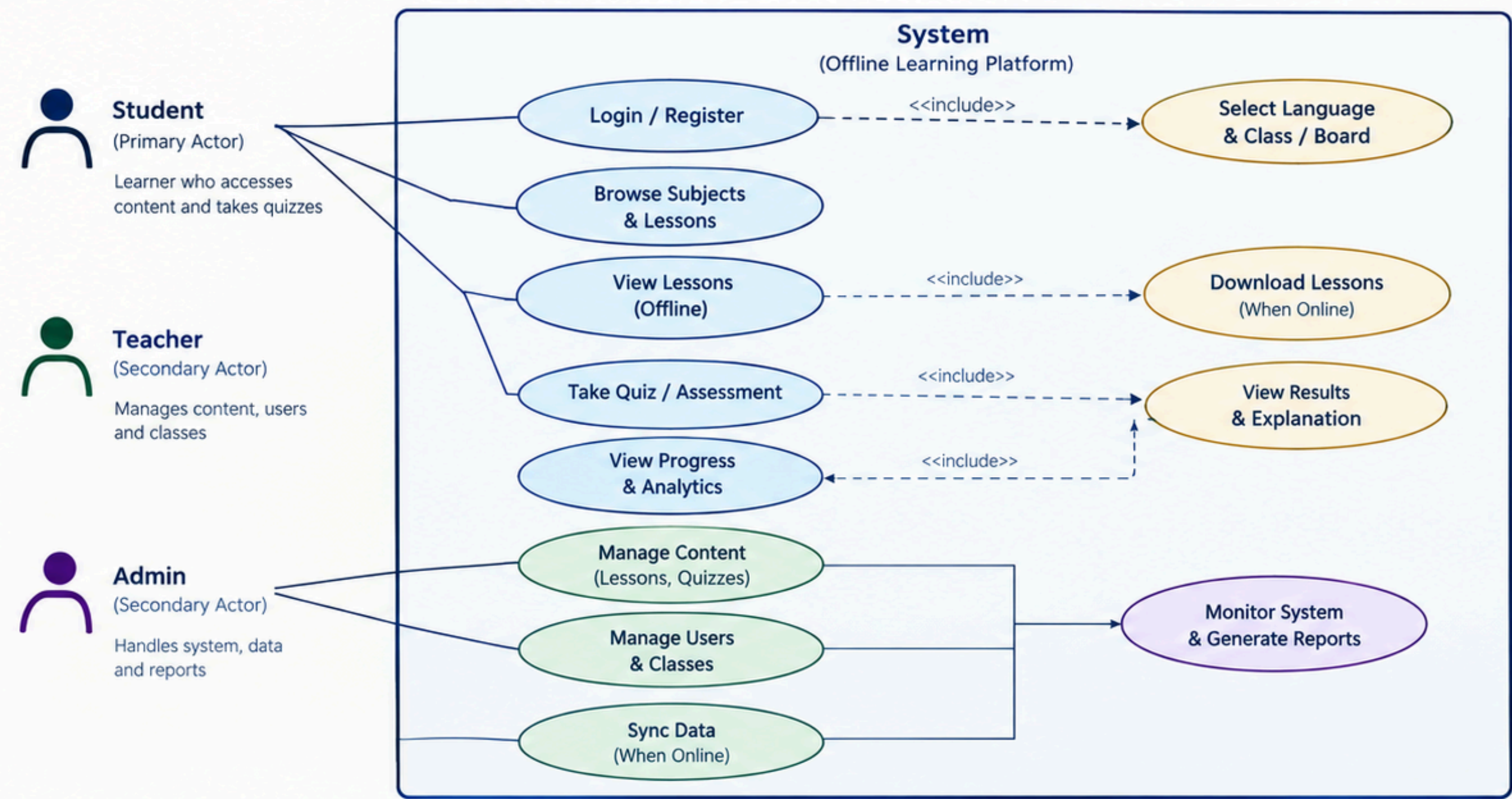
- Low-data mode
- Offline access
- Lightweight App
- Simple, student-friendly UI



LEARN • GROW • BRIGHTER FUTURES

OFFLINE LEARNING PLATFORM

USE CASE DIAGRAM



KEY FEATURES

-  **100% Offline Access**
Learn without internet, anytime, anywhere.
-  **Multi-Subject Learning**
Structured content for different classes & boards.
-  **Interactive Assessments**
Quizzes with instant results and explanations.
-  **Progress Tracking**
Personalized analytics for better learning outcomes.
-  **Role-Based Access**
Student, Teacher and Admin with dedicated controls.

IMPACT

 Bridging the digital divide	 Access to quality education	 Improved learning outcomes	 Empowered rural & underserved communities
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TECH STACK
HTML | CSS | JavaScript | IndexedDB | PWA

DATA FLOW
Local Storage → Offline Access → Sync (When Online)

OUR VISION
Education for Everyone, Everywhere.



OFFLINE LEARNING PLATFORM

SYSTEM ARCHITECTURE

Learn • Practice • Track • Grow — Without Internet

BRIDGING LEARNING
GAPS, OFFLINE



CLIENT APP

Android / Web App



Android / Web App

Seamless, responsive, cross-platform experience



Local Storage

Stores data for offline access



Offline Access

Works without internet



Background Sync

Syncs when connection is available

1

User & Setup

Login • Student ID + PIN



Login / Student ID + PIN



Language / Class / Board

2

Learning Content

Subjects & Lessons



Subjects & Lessons



Video / Audio / Notes



Download Content

3

Assessment & Progress

Quizzes



Quizzes



Results



Progress Tracking

4

Local Data & Sync

IndexedDB • Local Storage



IndexedDB
Local Storage



Sync Queue



Sync When Online

5

Device Services

Downloads • Notifications



Downloads



Notifications



Offline Cache



Use Offline



Save Locally



Reconnect



Sync with Server



BACKEND SERVICES

Secure • Scalable • Reliable



API Server

Handles requests & business logic



PostgreSQL Database

Stores user, content & progress data



Cloud File Storage

Videos, notes, downloadable content



Authentication

Secure login & authorization



Monitoring

Uptime, performance & error tracking



DESIGN GOAL

Reliable learning on
low-connectivity networks



10K+

Target Students

Accessible education for all



100%

Works Offline

No internet. No problem.



< 2s

Sync Time

Fast & seamless data sync



99.9%

Uptime (Cloud)

Reliable backend infrastructure

Learn Anywhere
Build a Brighter Future



OUR TECH STACK



NEXT.js

FRONTEND

Next.js

React framework for building fast, scalable and SEO-friendly web apps.

Express.js

BACKEND

Express.js

Fast, unopinionated backend framework for building RESTful APIs and web services.

Prisma

DATABASE (ORM)

Prisma ORM

Next-generation ORM for TypeScript & Node.js with type safety and automated migrations.

clerk

AUTHENTICATION

Clerk

Complete authentication solution with users, sessions, roles, permissions and social logins.



CLOUD SYNC

Firebase
Firestore



GAMIFICATION

Points, badges, levels & streaks



CONTENT

Downloadable videos, PDFs, quizzes & interactive lessons



AI (OPTIONAL)

Python + TensorFlow Lite for lightweight offline personalization



DEPLOYMENT

Android smartphones / tablets



Estimated Cost

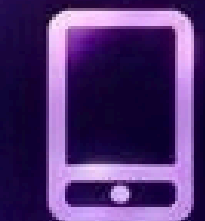


Estimated Cost (Project Prototype)



UI/UX & Learning
Materials

₹2,000



Testing on
Low-End Devices

₹3,000



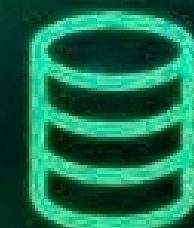
Cloud & Data Sync

₹2,000



App/Web
Development

₹5,000



Offline Database &
Storage

₹2,000



Local Language
Content

₹3,000





HACKVERSE

2026



THANK YOU